Online Appendix

Import Tariffs and the Systematic Response of Monetary Policy

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A Data sources

All data sources are described below. We use the arithmetic average across observations to aggregate to a quarterly frequency for all monthly data except the monetary shocks. In addition to the VAR data stated in the tables below, we use the FRED-QD data set (McCracken and Ng, 2021) and net-export consensus forecasts from the U.S. Survey of Professional Forecasters (SPF) to investigate invertibility concerns in Online Appendix D.1.

Table A.1: Import tariff measures

Variable	Data identifier and details
Trade-weighted average tariff rate	Computed as customs duties (*B235RC1Q027SBEA*) divided by dutiable imports. Dutiable imports are given by all goods imports (*A255RC1Q027SBEA*) multiplied by the average share of dutiable goods over all imported goods taken from the DataWeb of the United States International Trade Commission.
Unweighted average tariff rate	We use an approximate unweighted average tariff rate computed as follows. First, we compute all disaggregated trade-weighted tariff rates at the four-digit HTS-by-origin-country level. Then, we compute the (unweighted) arithmetic average across these disaggregated tariff rates. To limit the influence of small trading partners, we drop all countries with average import values below the median, where average import values are computed over the full sample. The data is retrieved from the DataWeb of the United States International Trade Commission.
Trade restrictiveness index	The index is provided by Schmitt-Grohé and Uribe (2025) and can be downloaded from the authors websites, e.g., here.

Notes: Data identifiers are in italic letters if taken from FRED, provided by the FRB of St. Louis.

Table A.2: Macroeconomic data

Variable	Data identifier and details
CPI inflation	*CPIAUCSL* and implied year-over-year inflation.
Core CPI inflation	*CPILFESL* and implied year-over-year inflation.
Inflation expectations	*MICH* (One-year horizon).
PPI inflation	*PPIACO* and implied year-over-year inflation.
Real GDP	*GDPC1.*
Real investment	*GPDIC1.*
Real consumption	*PCECC96.*

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Table A.2 (continued)

Variable	Data identifier and details
Imported consumption goods	Computed as nominal imported consumption goods (excluding automotive and food) (*A652RC1Q027SBEA*) divided by the GDP deflator (*GDPDEF*).
Real government spending	*GCEC1*.
Real exports	*EXPGSC1*.
Real imports	*IMPGSC1*.
Unemployment rate	*UNRATE*.
Industrial production	*INDPRO*.
Output gap	Computed as real GDP (*GDPC1*) divided by potential output (*GDPPOT*).
Terms of trade (non-petroleum goods)	*W371RG3Q020SBEA*.
Federal funds rate	*DFF*.
One-year Treasury yield	*GS1*.
Real federal debt	Computed as nominal federal debt (*FGSDODNS*) divided by the GDP deflator (*GDPDEF*).
Real government receipts	Computed as nominal federal receipts (*FGRECPT*) divided by the GDP deflator (*GDPDEF*) minus real customs duties (*B235RC1Q027SBEA*).
Macroeconomic uncertainty	We use the 12-month index, which can be downloaded here.
Shadow federal funds rate	We use the shadow federal funds rate from Wu and Xia (2016), which can be downloaded here. The series ends in 2023Q2 and we replace missing values by the effective federal funds rate.
Trade policy uncertainty	We use the quarterly index of Caldara, Iacoviello, Molligo, Prestipino, and Raffo (2020), which can be downloaded here.
Geopolitical risk	We use the monthly index of Caldara and Iacoviello (2022), which can be downloaded here.

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Table A.2 (continued)

Variable	Data identifier and details
Policy uncertainty	We use the 12 monthly categorical indices of Baker, Bloom, and Davis (2016) (overall economic policy uncertainty, monetary policy, fiscal policy, taxes, government spending, health care, national security, entitlement programs, regulation, financial regulation, trade policy, and sovereign debt and currency crises), which can be downloaded here. There is a single observation in which an aggregated index is zero (1991Q1 for sovereign debt and currency crises), which we replaced by the average of the two adjacent quarters before taking logarithms.
Romer and Romer (2004) shock	We use the extended shock that ends in 2007Q4, taken from McKay and Wolf (2023).
Miranda-Agrippino and Ricco (2021) shock	We use the shock that ends in 2014Q4, taken from McKay and Wolf (2023). They take the shock corresponding to the posterior mode of the reduced-form parameters of the original paper.
Jarociński (2024) shock	We use the conventional monetary shock (*u1*), which can be downloaded here. The series end in 2024Q3 and we set 2024Q4 to zero.
Hack, Istrefi, and Meier (2024)	We use the refined Romer and Romer (2004) shock taken from here.

Notes: The displayed variables are from FRED, provided by the FRB of St. Louis. Data identifiers are in italic letters. All remaining variables are provided by the mentioned scholars as indicated in the second column.

B Comparison with Schmitt-Grohé and Uribe (2025) and den Besten et al. (2026)

Schmitt-Grohé and Uribe (2025) estimate transitory and permanent U.S. import tariff shocks. Different from our results, they find that transitory import tariff shocks are deflationary and expansionary. Their permanent shocks yield a highly transitory inflation response and a mostly insignificant expansion of output. den Besten et al. (2026) estimate an annual VAR over 180 years and also find that tariffs are deflationary in the post-World War II subsample. However, in contrast to Schmitt-Grohé and Uribe (2025), they document contractionary effects of tariffs, robust across various subsamples. Below, we investigate why these results differ from our findings.

The core methodological differences between Schmitt-Grohé and Uribe (2025) and our approach are that

they use a state-space model, impose that the trade-weighted average tariff rate is exogenous to the economy (in their baseline), consider a different set of macroeconomic variables, and study a sample that starts earlier (1959Q2–2024Q4). In comparison with den Besten et al. (2026), there are similar differences in the macroeconomic variables included in the analysis, and their annual post-World War II subsample covers 1946–2024. In contrast to Schmitt-Grohé and Uribe (2025), they rely on more conventional time series techniques and a narratively identified external instrument. The latter is based on 10 events that are classified as exogenous in the aforementioned sample.

To compare with their results, we use our baseline identification approach and focus on the sample period. First, we re-estimate our VAR but let our sample start in 1980Q1 to ensure that our findings are not unduly sensitive to 1990Q1 as sample start. The results are shown as solid blue lines in Figure B.1. In this sample, our main findings remain unchanged. In particular, we still find that tariff shocks are inflationary and contractionary.

Next, we go further and start the sample in 1967Q1, which is the earliest feasible sample start due to data availability.[1] The results are shown as gray dashed-dotted lines. Indeed, in this extended sample, we find that the shock is deflationary and not contractionary anymore, broadly consistent with Schmitt-Grohé and Uribe (2025).

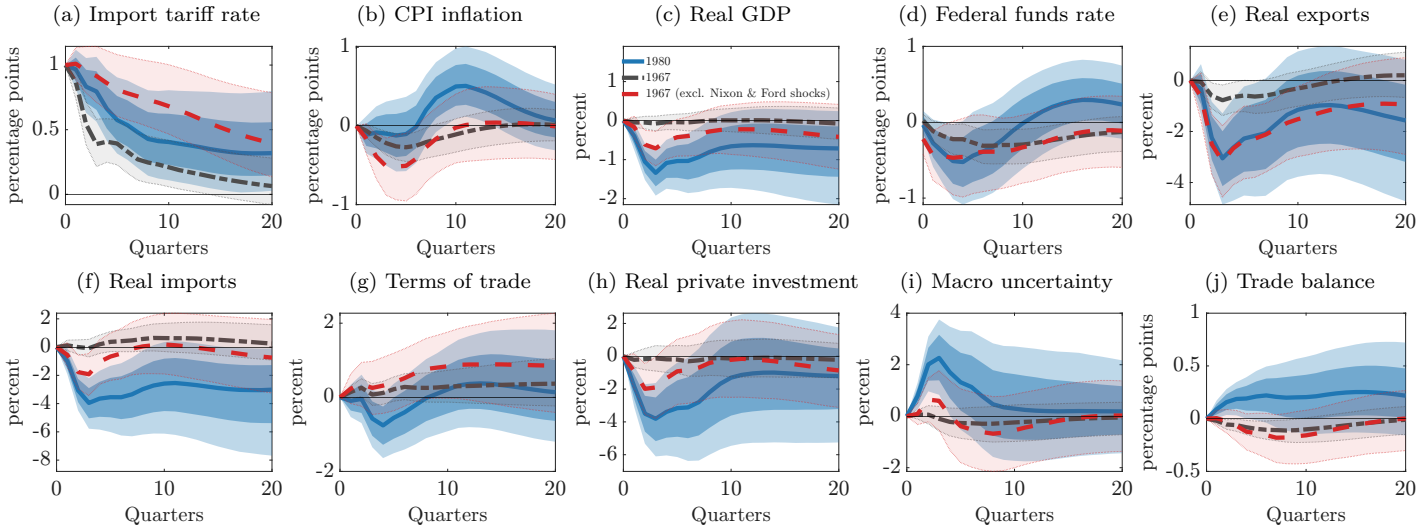
However, the extended sample includes two large tariff spikes, the well-known Nixon and Ford tariff events, which are also discussed in Schmitt-Grohé and Uribe (2025); see their Figure 4. The Nixon tariffs partly capture the endogenous response of the US government to the turmoil associated with the end of the Bretton Woods system. The Ford tariffs were levied on oil imports, partly in response to high oil prices. Thus, both of these events are classified as endogenous in the narrative account of den Besten et al. (2026).

Thus, we investigate how the results change if we remove these two tariff spikes from the raw series. Specifically, we smooth out these events from the average tariff rate time series in the following way: We replace the values during these episodes with the average import tariff across the pre- and post-event tariff rate.[2] The responses using the smoothed import tariff series are shown as dashed red lines in Figure B.1. It turns out that we recover a significant output contraction, which is also visible in other components of aggregate demand, e.g., in real private investment. Therefore, we conclude that there is robust evidence for tariffs being contractionary, consistent with den Besten et al. (2026), while the effects on inflation are more sample dependent. A potential reason for this sample dependence is that supply chain length and complexity have increased substantially, so that more products are affected by tariff changes in recent decades. In a similar vein, Bergin and Corsetti (2023) and Moro and Landi (2026) show that a sufficiently low share of imported intermediate inputs in marginal cost may predict tariff hikes to be deflationary. Investigating such

[1] An earlier sample start is infeasible since the terms of trade index is only available starting in 1967.

[2] The Nixon tariff event took place in the third and fourth quarters of 1971. The Ford tariff event took place during all quarters of 1975.

Figure B.1: Varying sample starts

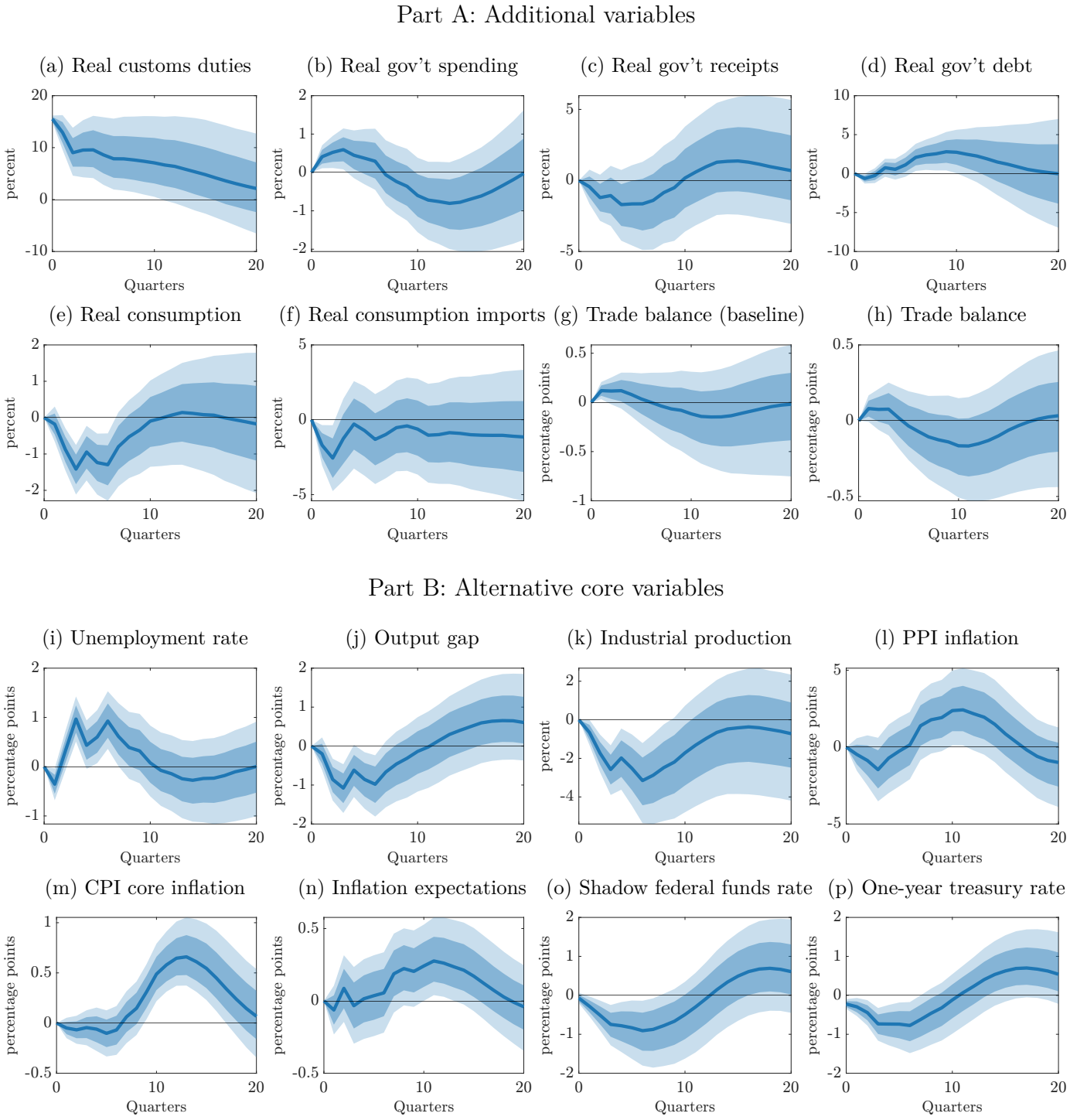


Notes: The solid blue, dashed-dotted gray, and dashed red lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted lines indicate 90% credible sets. The legend indicates the sample start of the respective specification; see text for details.

dependencies in greater detail is left for future research.

C Responses of additional outcome variables

Figure C.1: Responses of further macroeconomic variables to the import tariff shock



Notes: This figure shows impulse responses estimated based on a Bayesian VAR. The solid blue line represents the posterior median, and the shaded areas are 68% and 90% credible sets. In Part A, Panels (a)-(f), we augment the baseline VAR by each variable individually and re-estimate the model. We impose a zero impact effect on all additional variables, except for customs duties. In Panel (g), we use our baseline VAR and compute the implied trade balance response, see Footnote 23 in the main text for details. In Panel (h), we include the trade balance relative to GDP instead of real imports and exports in the VAR. In Part B, we replace individual variables from the baseline VAR by alternative measures. We replace real GDP by alternative measures of real activity in Panel (i) to (k). We replace CPI inflation by other price measures in Panel (l) to (n). We replace the federal funds rate by alternative interest rates in Panel (o) to (p).

D Sensitivity analysis

D.1 Testing partial invertibility

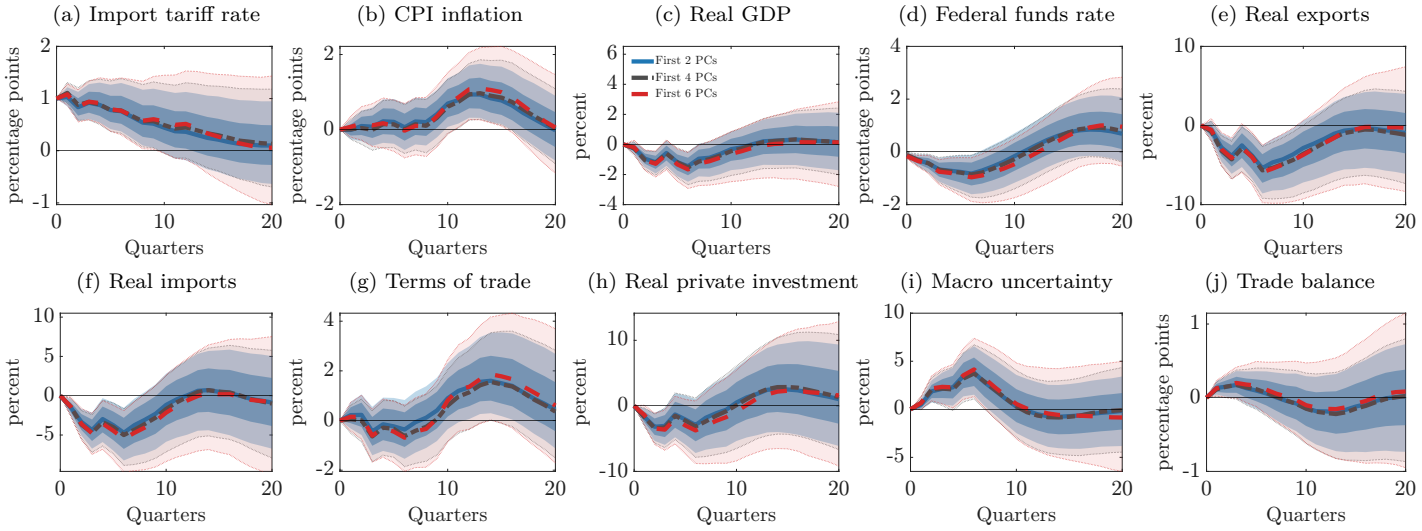
To ensure that our baseline VAR contains sufficient information to identify the tariff shock, we perform the orthogonality test proposed by Forni and Gambetti (2014). First, we estimate the principal components using a large quarterly dataset of macroeconomic and financial variables for the U.S. economy, taken from McCracken and Ng (2021). Then, we test for orthogonality with respect to the lags of these principal components. To select the optimal number of principal components to be included in the test, we rely on the criteria proposed by Alessi, Barigozzi, and Capasso (2010), which suggests the first four or first six principal components. The results of the orthogonality test are reported in Table D.1. In all cases, we cannot reject the null hypothesis of orthogonality, indicating a lack of shock predictability by the information set spanned by the principal components. As an additional test, we include the first two, four, and six principal components in the VAR and order them after the federal funds rate. The results in Figure D.1 are similar to the baseline. This indicates that our results are not driven by the omission of important macroeconomic and financial variables from our VAR. Overall, both exercises indicate that our baseline VAR is informationally sufficient to identify the tariff shock.

Table D.1: Testing partial invertibility based on Forni and Gambetti (2014)

	First 4 PCs, k lags				First 6 PCs, k lags			
	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$
F-stat	0.187	0.392	0.412	0.438	0.241	0.466	0.526	0.479
p-value	0.945	0.923	0.957	0.969	0.962	0.931	0.941	0.980

Notes: The table shows the results of the "Structuralness" test proposed in Forni and Gambetti (2014). Specifically, we report the F-statistics and the p-values for regressions of the tariff shock on up to k lags of the first four and six principal components. The shock is based on the posterior mean of the Tariff VAR, taken over the reduced-form parameters.

Figure D.1: Include principal components in VAR vector



Notes: The solid blue, dashed-dotted gray, and dashed red lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted lines indicate 90% credible sets. We include principal components in the VAR and allow the tariff shock to impact these principal components in the quarter of the shock; see text for details.

A possible caveat of the above approach is that the McCracken and Ng (2021) data is not specific to tariffs, and its variation could be dominated by non-tariff shocks. Thus, we present another predictability test using forecasts of future trade flows from the Survey of Professional Forecasters (SPF). The core idea is that SPF participants should incorporate the effect of a tariff shock once it is announced. If our identified tariff shock picks up these anticipated events, it follows that it should be predictable by the SPF forecasts. Therefore, we estimate

$$\varepsilon_t^\tau = \alpha + \sum_{h=0}^{3} \beta^h E_{t-1}\left[NX_{t+h} - NX_{t-1}\right] + v_t, \tag{D.1}$$

where ε_t^τ is our identified tariff shock (based on the posterior mean of the reduced-form parameters), and the regressors $E_{t-1}\left[NX_{t+h} - NX_{t-1}\right]$ denote the date $t-1$ expected change in net exports over the next $h+1$ periods, based on the consensus forecast. v_t is an error term.[3] Table D.2 shows the point estimates of β^h and associated standard errors in parentheses, which are robust to heteroskedasticity and serial correlation. The results indicate that our tariff shock is not predictable. All coefficients are statistically insignificant at conventional levels and the R^2 is fairly low. This holds regardless of whether we estimate these effects individually or jointly. Therefore, we conclude that anticipation (and subsequent invertibility failures) likely has a negligible effect on our estimated responses.

[3]We opted for this specification because we find it particularly intuitive. When using exactly the same regression specification as for the McCracken and Ng (2021) data, we obtain similar results.

Table D.2: Tariff shock and SPF net-export expectations

	(1)	(2)	(3)	(4)	(5)
$E_{t-1}[NX_t - NX_{t-1}]$	-0.0139				-0.0327
	(0.0120)				(0.0343)
$E_{t-1}[NX_{t+1} - NX_{t-1}]$		-0.0066			-0.0214
		(0.0075)			(0.0379)
$E_{t-1}[NX_{t+2} - NX_{t-1}]$			-0.0040		0.0651
			(0.0054)		(0.0481)
$E_{t-1}[NX_{t+3} - NX_{t-1}]$				-0.0034	-0.0341
				(0.0044)	(0.0270)
R^2	0.014	0.009	0.006	0.007	0.031
Observations	136	136	136	136	136

Notes: The table shows regression results based on equation (D.1). The shock is based on the posterior mean of the Tariff VAR, taken over the reduced-form parameters. Standard errors robust to heteroskedasticity and serial correlation are shown in parentheses, with corresponding asterisks for statistical significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

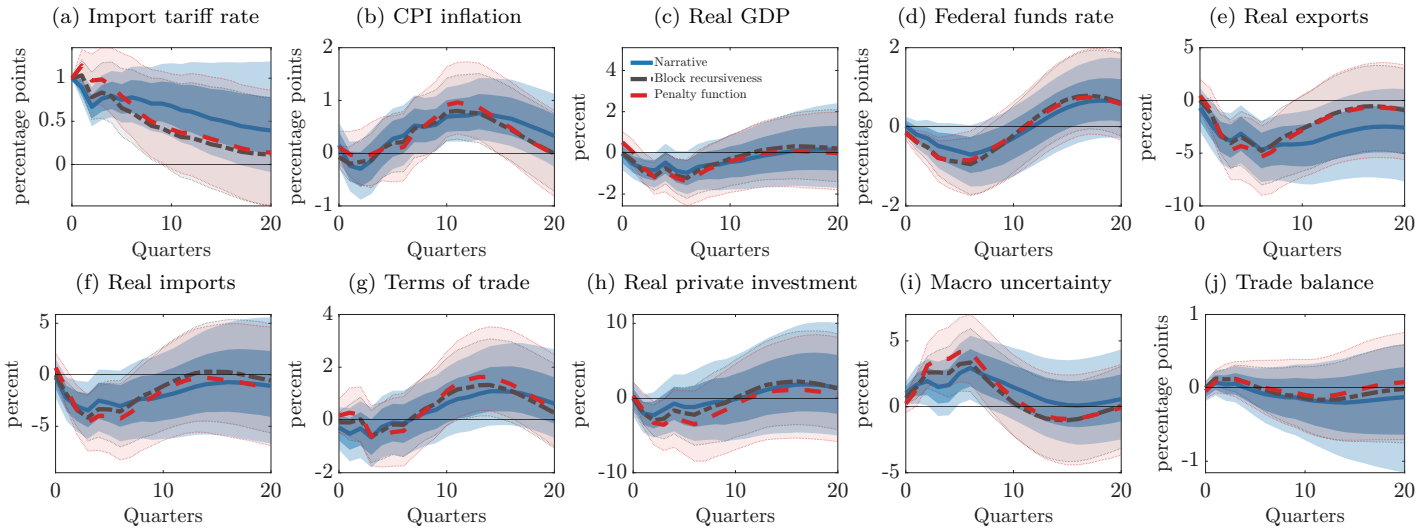
D.2 Alternative identification

Narrative identification Our baseline tariff rate measure could pick up variation unrelated to tariff policy changes. To address the concern that our estimates pick up these alternative sources of variation, we compile a series of narratively identified tariff policy changes, which are unlikely to be confounded by the above-described issues. To this end, we define an indicator, $\mathbb{I}_t\{\text{tariff policy event}\} \in \{0, 1\}$, which is only activated when we identify a tariff policy change in quarter t. Table D.3 lists all of these events. Finally, we compute our narrative tariff change time series as $\Delta\tau_t^{narr} = \mathbb{I}_t\{\text{tariff policy event}\}\Delta\tau_t$, where $\Delta\tau_t$ denotes the quarter-on-quarter change in the trade-weighted average tariff rate. We include this series as an exogenous variable in the VAR.

Table D.3: Narratively identified tariff policy changes

#	Event description	Quarterly date
(1)	NAFTA Agreement	1994Q1
(2)	WTO Establishment	1995Q1
(3)	US–Canada SLA Agreement I	1996Q2
(4)	WTO ITA I	1997Q3
(5)	Trade & Development Act (Sub-Saharan Africa)	2000Q2
(6)	Bush Steel Safeguard	2002Q1
(7)	Early Removal of Bush Steel Safeguard	2003Q4
(8)	US–Canada SLA Agreement II	2006Q4
(9)	WTO ITA II	2016Q3
(10)	US Tariffs (Solar, Washing Machines, Steel, Aluminum)	2018Q1
(11)	Section 232 Expansion (EU, Canada, Mexico)	2018Q2
(12)	Section 301 China Tariffs – List 1 & 2	2018Q3
(13)	Section 301 China Tariffs – List 3	2018Q4
(14)	Section 301 China Tariffs – List 3 Increase	2019Q1

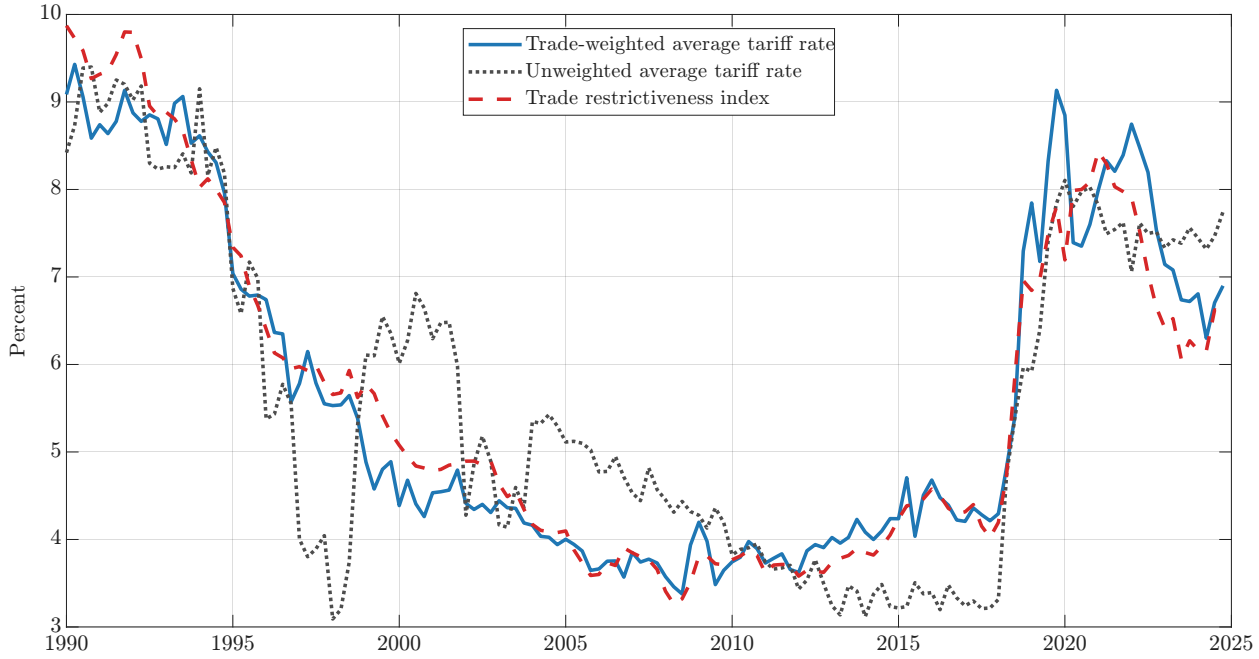
Figure D.2: Alternative identification approaches



Notes: The solid blue, dashed-dotted gray, and dashed red lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted lines indicate 90% credible sets. The legend indicates the identification approach of the respective specification; see main text for details.

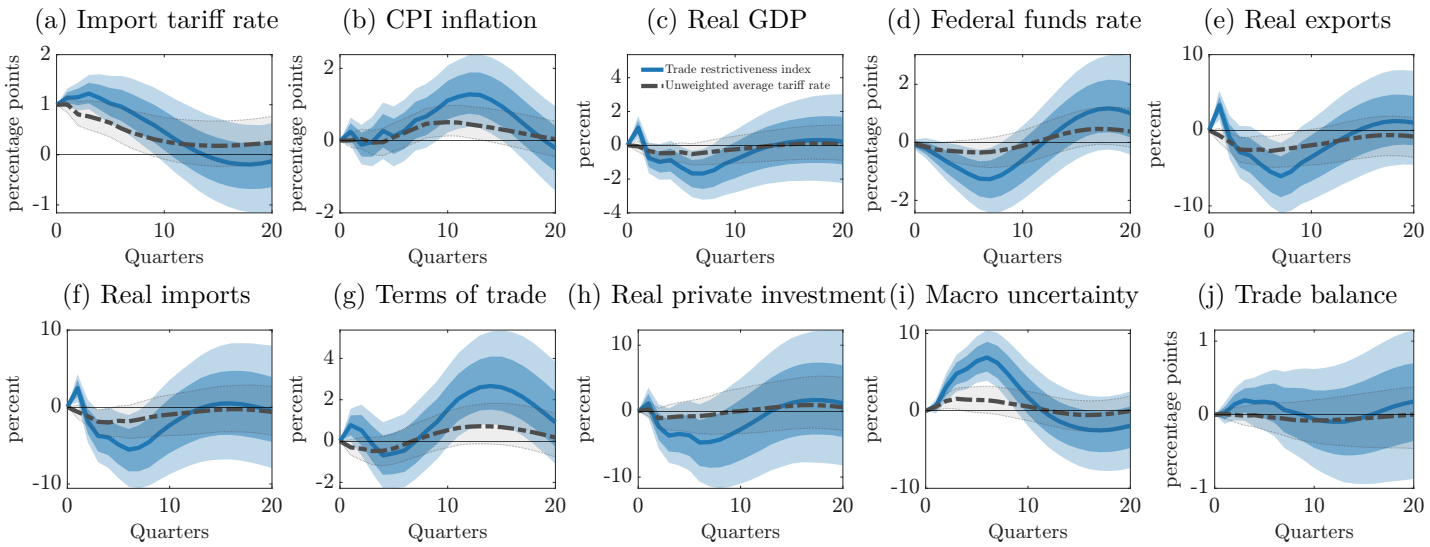
D.3 Alternative tariff measurement

Figure D.3: Comparison across tariff measures



Notes: This figure shows the time series of the (i) trade-weighted and (ii) unweighted average U.S. import tariff rates, as well as (iii) the trade restrictiveness index from Schmitt-Grohé and Uribe (2025). Series (ii) and (iii) have been standardized to match the mean and standard deviation of series (i) to ease comparison.

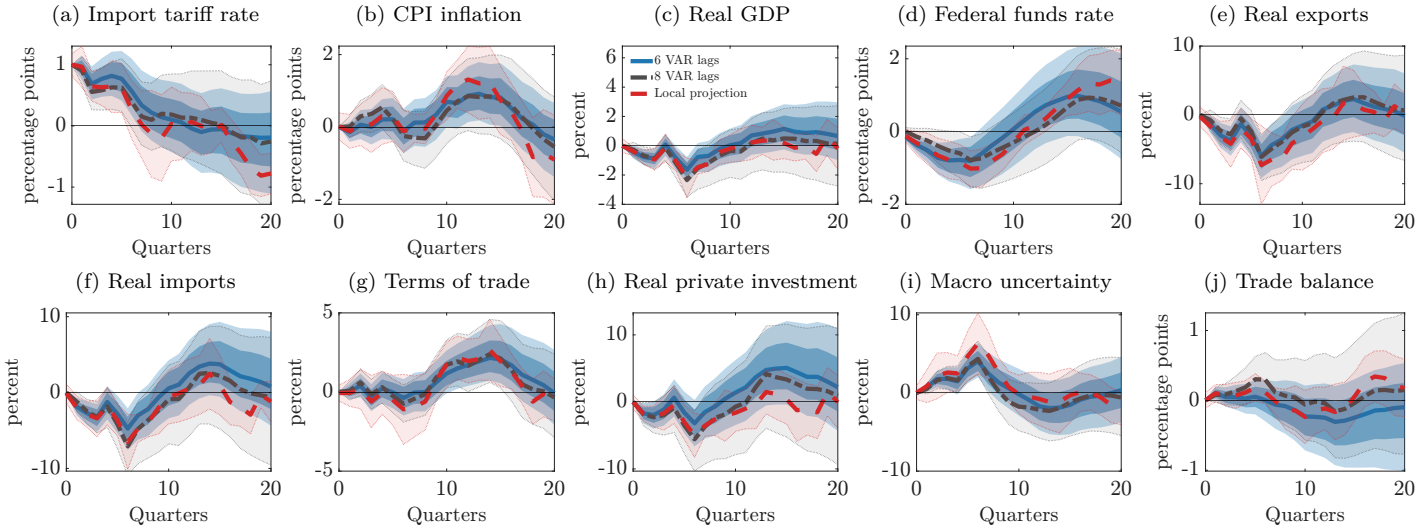
Figure D.4: Responses to the import tariff shock using alternative tariff measures



Notes: The solid blue and dashed-dotted gray lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted lines indicate 90% credible sets. The legend indicates the tariff measure used in the respective specification; see main text for details. We normalize both tariff series to have the same variance as the trade-weighted average tariff rate to make them as comparable as possible.

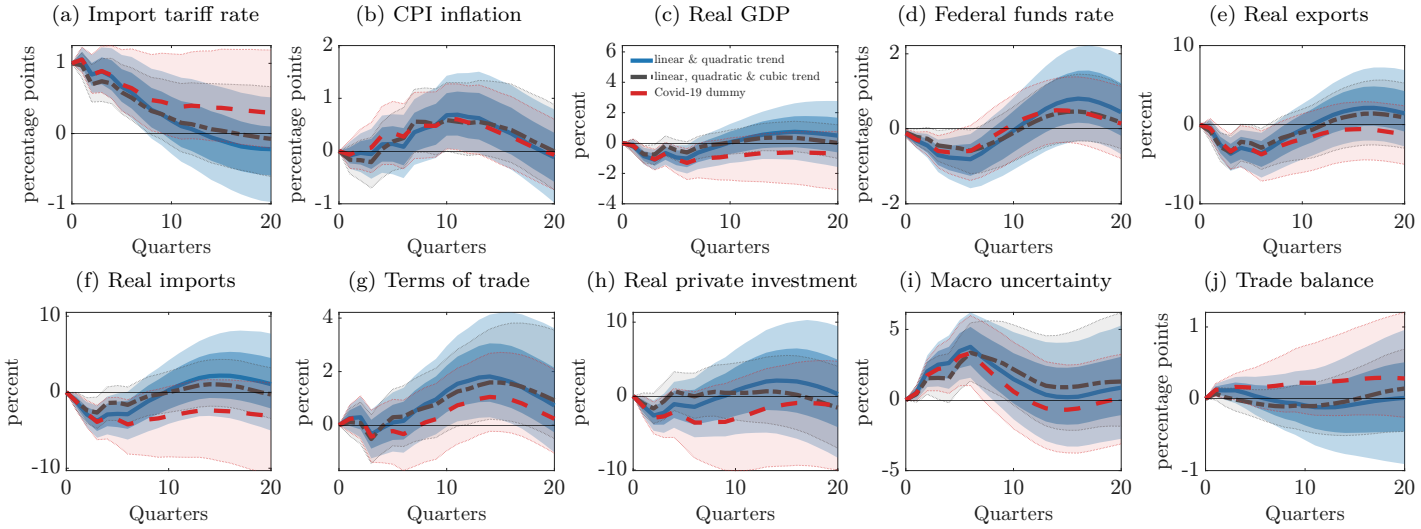
D.4 Further sensitivity analysis

Figure D.5: Relaxing VAR assumptions



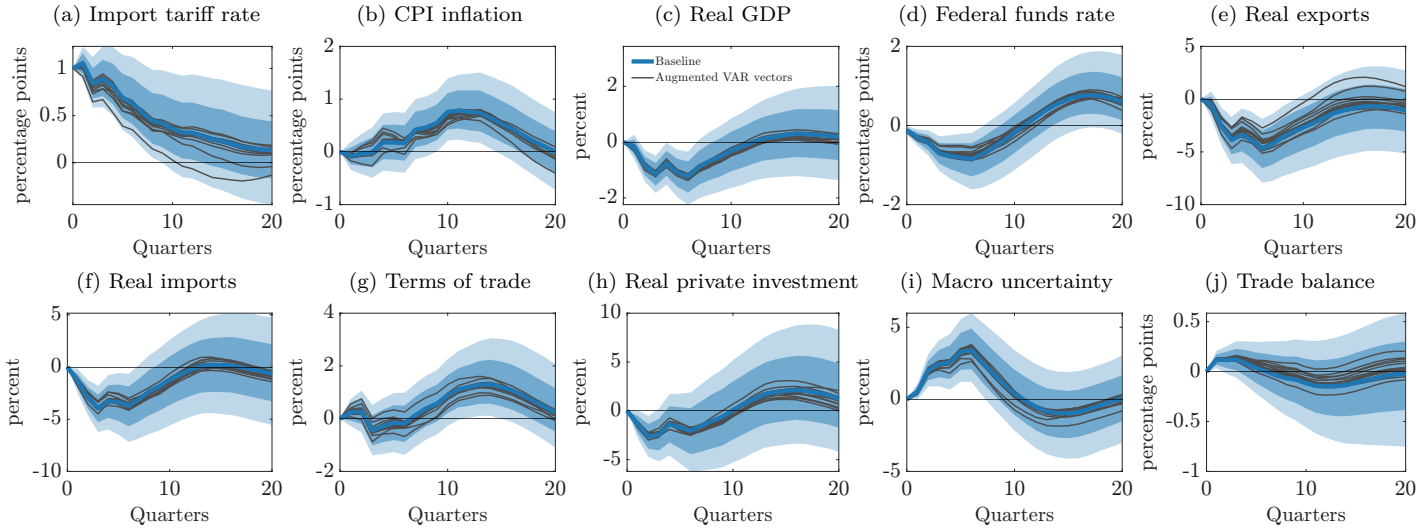
Notes: The solid blue and dashed-dotted gray lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted gray lines indicate 90% credible sets. The dashed red line corresponds to a local projection point estimate, and the thin-dotted red lines indicate 90% confidence bands based on standard errors robust to heteroskedasticity and serial correlation. The local projection includes four lags of the shock and the outcome variable as control variables, and a linear time trend. The shock is based on the posterior mean of the baseline tariff VAR, taken over the reduced-form parameters.

Figure D.6: Additional deterministic variables



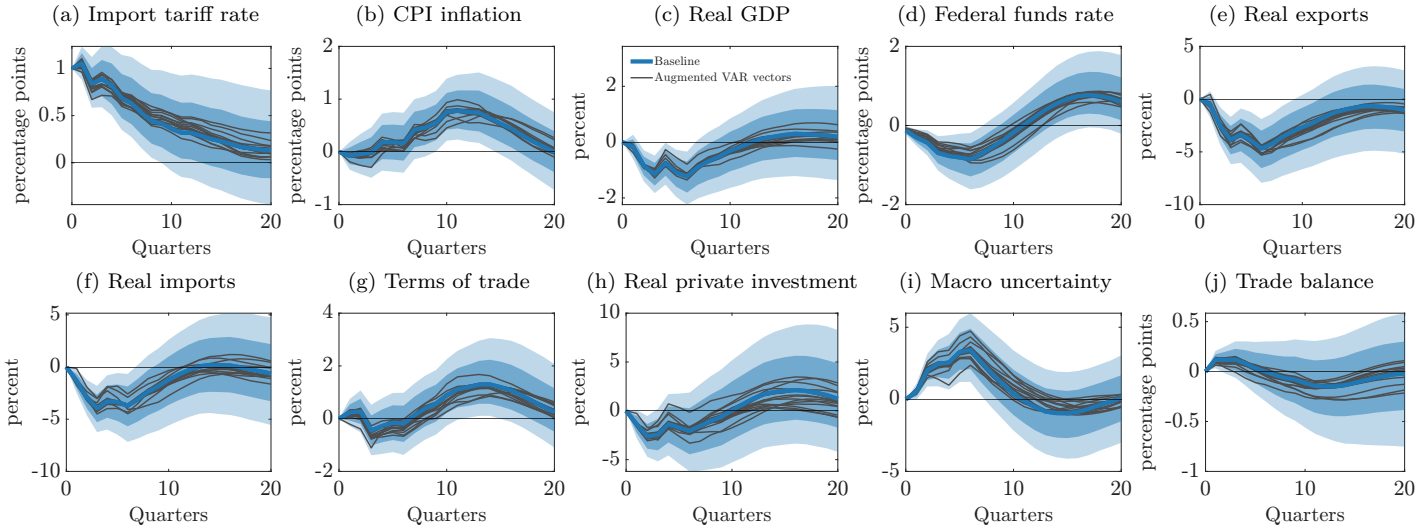
Notes: The solid blue, dashed-dotted gray, and dashed red lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted lines indicate 90% credible sets. The legend indicates the included deterministic variables of the respective specification.

Figure D.7: Controlling for additional measures of uncertainty



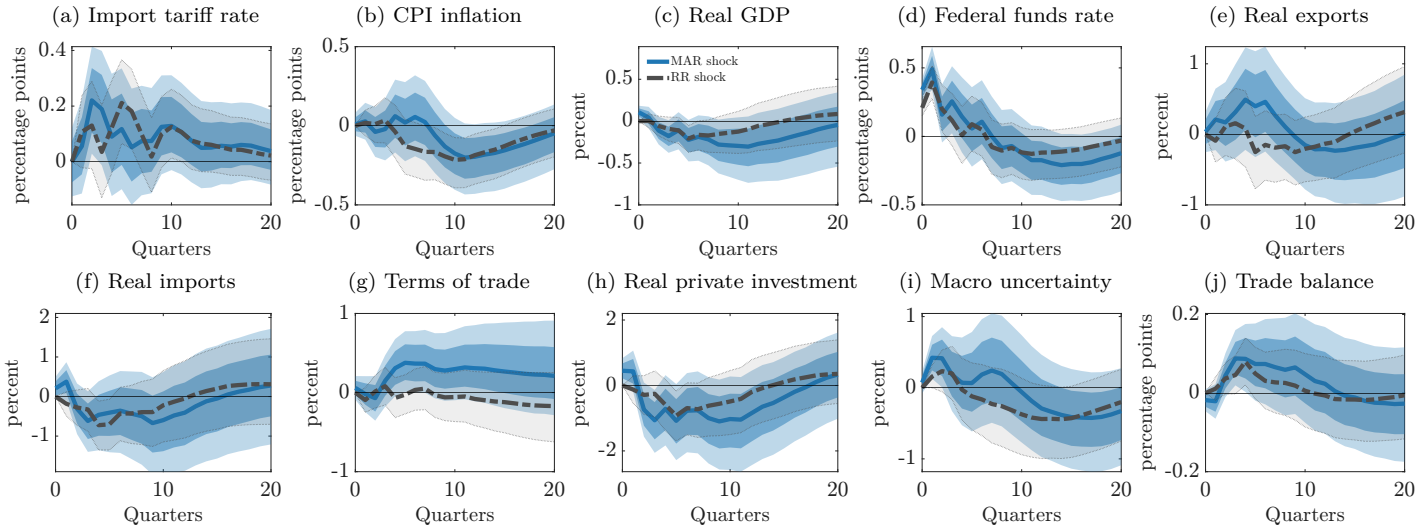
Notes: The solid blue lines represent the posterior median of the baseline tariff VAR, and the blue shaded areas are 68% and 90% credible sets. Each thin gray line corresponds to the posterior median of an augmented VAR, where we include an additional measure of uncertainty and allow the tariff shock to impact this variable in the quarter of the shock. We consider the 12 categorical economic policy uncertainty indices from Baker et al. (2016), the geopolitical risk measure from Caldara and Iacoviello (2022), and the trade policy uncertainty index from Caldara et al. (2020). All indices are available via the website https://www.policyuncertainty.com.

Figure D.8: Controlling for additional variables from Figure C.1



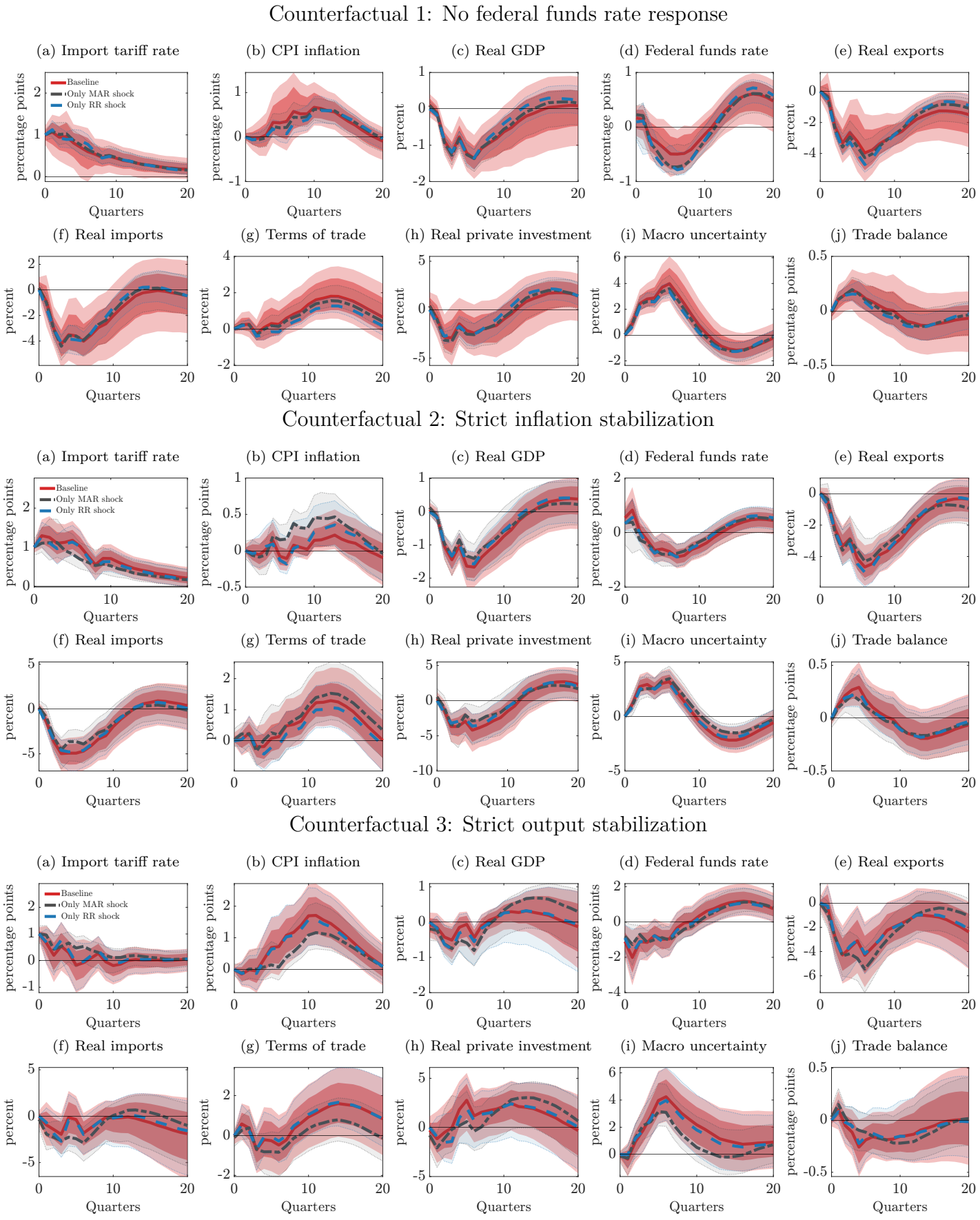
Notes: The solid blue lines represent the posterior median of the baseline tariff VAR, and the blue shaded areas are 68% and 90% credible sets. Each thin gray line corresponds to the posterior median of an augmented VAR, where we include an additional variable to the VAR or replace a baseline measure by a different variable. The displayed specifications correspond to all VAR results shown in Figure C.1.

Figure D.9: Baseline responses to monetary policy shocks



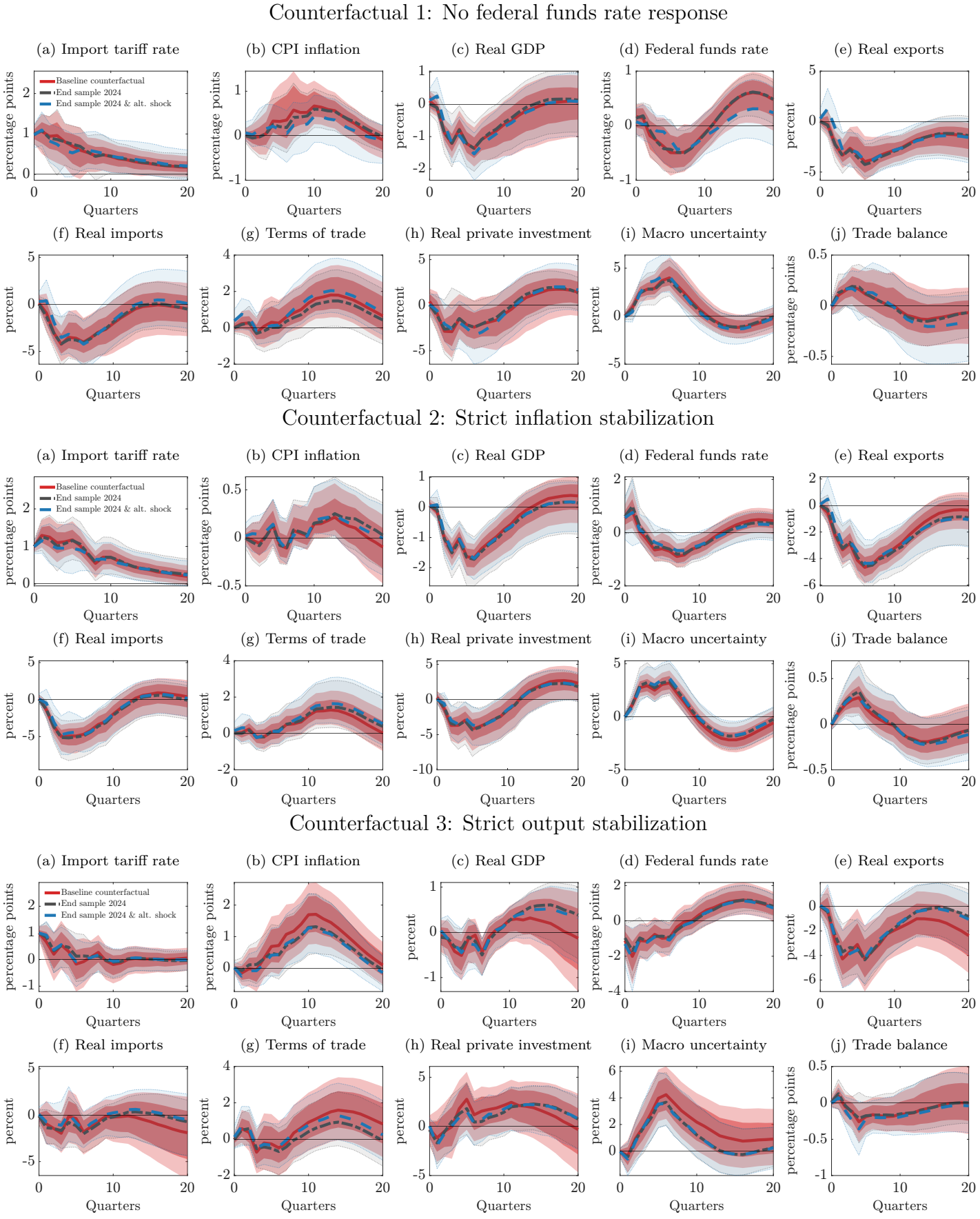
Notes: The solid blue and dashed-dotted gray lines represent posterior medians. The blue shaded areas are 68% and 90% credible sets, and the thin-dotted gray lines indicate 90% credible sets. The results are from the baseline monetary policy VAR.

Figure D.10: Counterfactuals using only a single monetary policy shock



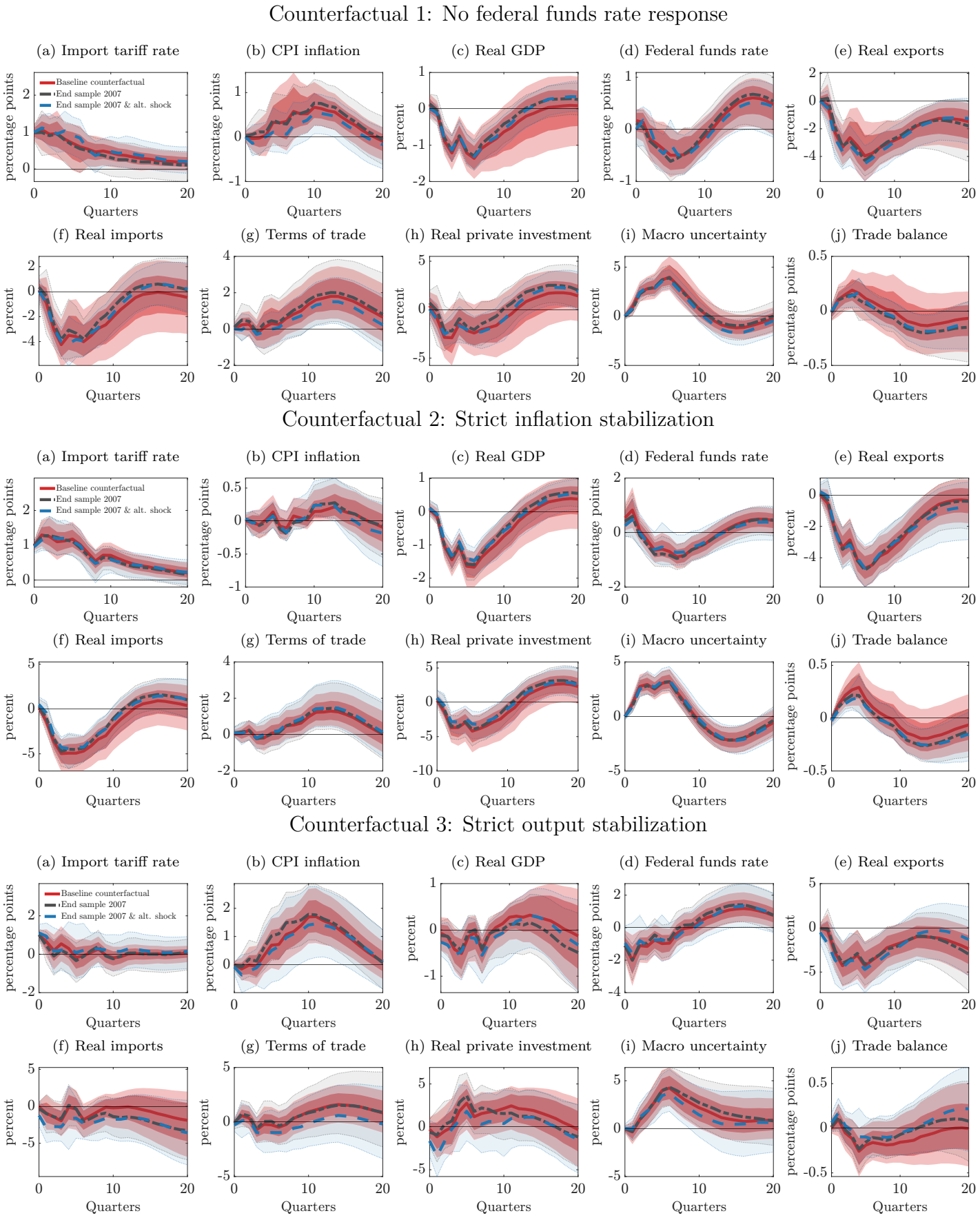
Notes: Alternative monetary policy counterfactuals in comparison to the baseline counterfactual.

Figure D.11: Counterfactuals using an extended sample and the Jarociński (2024) shock



Notes: Alternative monetary policy counterfactuals in comparison to the baseline counterfactual. Extended sample end to 2024 and Jarociński (2024) *u1* shock instead of MAR.

Figure D.12: Counterfactuals using an shorter sample and the Hack et al. (2024) shock



Notes: Alternative monetary policy counterfactuals in comparison to the baseline counterfactual. Using a shorter sample and Hack et al. (2024) shock instead of RR.

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